

Chapter 2: Discrete-Time Stochastic Optimal Control

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Abstract. *This chapter considers optimal policy synthesis for discrete-time stochastic optimal control problems. We first study the dynamic programming algorithm for state feedback synthesis, followed by state estimation via Bayesian filtering. Finally, we introduce the belief state representation of partially observable stochastic optimal control problems, and discuss output feedback synthesis via dynamic programming over belief states.*

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2.1 Introduction

The previous chapter introduced several formulations of stochastic optimal control problems. Needless to say, however, formulating the problem is not our ultimate goal – our goal is to solve it!

By “solving” a stochastic optimal control problem, we mean obtaining an algorithm that can sample u_t from the optimal control policy $\mu_t(du_t|h_t)$ for all possible realizations of the input-output history $h_t = (y_{0:t}, u_{0:t-1})$. To be useful in practice, this algorithm should be computationally efficient, allowing u_t to be generated within a reasonable time frame. If efficient sampling from the optimal policy is infeasible, we must consider sampling from a “near-optimal” policy, which we hope is still useful in practice. In such cases, it is important to quantify the notion of “near-optimality” to provide performance guarantees for the proposed solution.

With these goals in mind, this chapter presents a general recipe for solving stochastic optimal control problems. The central concept that we utilize for this development is *Bellman's optimality principle* [1]:

An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.

Since Bellman's optimality principle characterizes the optimal policy as a function of states, it is most naturally applied to stochastic optimal control problems with fully observable states. Therefore, we begin this chapter by addressing state feedback synthesis for fully observable stochastic optimal control problems. We will see that Bellman's optimality principle leads to computationally tractable solutions in important special cases, including finite Markov Decision Processes (MDPs) and stochastic Linear-Quadratic Regulator (LQR) problems.

Next, we turn our attention to partially observable systems. In particular, we examine the problem of *Bayesian filtering*, where the goal is to compute the posterior probability distribution of the hidden state (often referred to as the *belief*) given the history of measurements. We will see that Bayesian filtering is computationally tractable in both the finite-state case and the linear-Gaussian case. The former is known as the hidden Markov model (HMM) filter, and the latter as the Kalman filter.

We then pursue the idea of combining these two components – state feedback synthesis and Bayesian filtering – to address control of partially observable systems. Specifically, we introduce the so-called *belief-state formalism*, a systematic approach for converting a stochastic optimal control problem with partially observable states into one with fully observable states. The key idea is to treat the belief itself as the state of the underlying optimal control problem. We will see that this approach is always valid and can sometimes lead to computationally tractable solutions for partially observable stochastic optimal control problems. A well-known example is the Linear-Quadratic-Gaussian (LQG) control problem, which can be solved by a simple combination of LQR and Kalman filtering.

In general, however, the belief-state formalism only offers conceptual benefits: applying Bellman's optimality principle directly in belief space typically does not yield computationally efficient algorithms. In Section 2.4.2, we will see that the tractability of fully observable MDPs and HMM filtering does not imply the tractability of POMDPs.¹ Nevertheless, the belief-state formalism remains a foundation for developing powerful suboptimal solution methods for POMDPs, including *fully observable value approximation* [2, 3] and *point-based value iteration* [4].

2.2 Optimal State Feedback Synthesis

Let us first focus on the fully observable state-space model $\tau_t(dx_{t+1}|x_t, u_t)$ with the initial state x_0 , and consider the finite-horizon stochastic optimal control problem:

$$\min_{\mu_{0:T-1}} \mathbb{E}^{\mu_{0:T-1}} \left\{ \sum_{t=0}^{T-1} C_t(x_t, u_t) + C_T^{\text{exit}}(x_T) \right\}. \quad (2.1)$$

¹It has been shown that solving POMDPs is a PSPACE-hard problem.

Note that (2.1) is a minimization problem over the space of history-dependent randomized policies $\mu_{0:T-1} = \{\mu_t(du_t|h_t)\}_{t=0}^{T-1}$, where $h_t = (x_{0:t}, u_{0:t-1})$. The notation $\mathbb{E}^{\mu_{0:T-1}}$ indicates that the expectation is taken with respect to the joint probability measure induced by the chosen policy $\mu_{0:T-1}$. The optimal solution that attains the minimum may or may not exist, depending on the structure of the underlying state-space model and the cost functional. Therefore, we begin by focusing on a simple case in which the existence of an optimal solution is guaranteed.

2.2.1 Finite MDP Case

Let us assume that both $\mathcal{X}$ and $\mathcal{U}$ are finite sets and show that, in this case, there exists an optimal policy that attains the minimum in (2.1). To this end, we derive a sufficient condition for optimality and show that a policy satisfying this condition exists. The condition we derive below corresponds to what is known as *Bellman's optimality criterion*, which is expressed in terms of a system of equations commonly referred to as *Bellman's equation* or the *dynamic programming equation*. Conversely, we also show that any optimal policy necessarily satisfies Bellman's equation.

Let $\mu_{0:T-1} = \{\mu_t(u_t|h_t)\}_{t=0}^{T-1}$ be a fixed (not necessarily optimal) policy. Consider the problem of evaluating the cost functional in (2.1). Using its stage-additive structure, this evaluation can be performed recursively in a backward-in-time manner.

To describe this computational procedure, let $h_t = (x_{0:t}, u_{0:t-1})$ be the history of state-input pairs up to time t . For each t and h_t , define the *cost-to-go function* as

$$J_t^\mu(h_t) := \mathbb{E}^{\mu_{t:T-1}} \left\{ \sum_{k=t}^{T-1} C_k(x_k, u_k) + C_T^{\text{exit}}(x_T) \right\}. \quad (2.2)$$

Notice that (2.2) represents the expected future cost incurred by following the policy $\mu_{t:T-1}$ from time t onward. At the terminal step, we define the cost-to-go function as the terminal cost:

$$J_T^\mu(h_T) := C_T^{\text{exit}}(x_T). \quad (2.3)$$

Now observe that $J_t^\mu(\cdot)$ can be recursively expressed in terms of $J_{t+1}^\mu(\cdot)$ as follows:

$$J_t^\mu(h_t) = \sum_{u_t \in \mathcal{U}} \left[C_t(x_t, u_t) + \sum_{x_{t+1} \in \mathcal{X}} J_{t+1}^\mu(h_{t+1}) \tau(x_{t+1}|x_t, u_t) \right] \mu_t(u_t|h_t). \quad (2.4)$$

In (2.4), the history $h_{t+1} = (h_t, u_t, x_{t+1})$ is formed by appending to h_t the new components u_t and x_{t+1} , which are generated by the policy $\mu_t(u_t|h_t)$ and the state transition kernel $\tau_t(x_{t+1}|x_t, u_t)$, respectively.

Equation (2.4) together with the terminal condition (2.3) allows us to compute the values of the cost-to-go functions backward in time. Once the algorithm terminates, we obtain the cost-to-go function $J_0^\mu(x_0)$, which is nothing but the value of the cost function (2.1).

Next, let us consider the problem of choosing the optimal policy $\mu_{0:T-1}$ that minimizes the cost function (2.1). Does there always exist an optimal policy? If so, how can we find such a policy? The next theorem answers these questions.

Theorem 2.1 Consider the stochastic optimal control problem (2.1) where $\mathcal{X}$ and $\mathcal{U}$ are finite spaces. Consider the following backward recursion:

$$J_t(h_t) = \min_{\mu_t(u_t|h_t)} \sum_{u_t \in \mathcal{U}} \left[C_t(x_t, u_t) + \sum_{x_{t+1} \in \mathcal{X}} J_{t+1}(h_{t+1}) \tau_t(x_{t+1}|x_t, u_t) \right] \mu_t(u_t|h_t) \quad (2.5)$$

for $t = T-1, T-2, \dots, 1, 0$ with the terminal condition $J_T(h_T) = C_T^{\text{exit}}(x_T)$.

- (i) For all $t = 0, 1, \dots, T-1$ and for all h_t , there exists an optimal solution $\mu_t^*(u_t|h_t)$ that attains the minimum of (2.5).
- (ii) Let $\mu_{0:T-1}^* = \{\mu_t^*(u_t|h_t)\}_{t=0}^{T-1}$ be the policy formed by $\mu_t^*(u_t|h_t)$, $t = 0, 1, \dots, T-1$ obtained in part (i). Then, the cost-to-go function associated with the policy $\mu_{0:T-1}^*$ coincides with the solution $J_t(h_t)$ to the backward recursion (2.5). That is,

$$J_t(h_t) = J_t^{\mu^*}(h_t) \quad (2.6)$$

holds for all $t = 0, 1, \dots, T$ and h_t .

- (iii) The policy $\mu_{0:T-1}^*$ constructed in part (ii) minimizes the cost-to-go function, i.e.,

$$J_t^{\mu^*}(h_t) = \min_{\mu_{t:T-1}} \mathbb{E}^{\mu_{t:T-1}} \left[\sum_{k=t}^{T-1} C_k(x_k, u_k) + C_T^{\text{exit}}(x_T) \right] \quad (2.7)$$

holds for all $t = 0, 1, \dots, T-1$ and h_t .

- (iv) The policy $\mu_{0:T-1}^*$ constructed in part (ii) is an optimal solution to the stochastic optimal control problem (2.1). Moreover, the solution $J_0(x_0)$ obtained by the backward recursion (2.5) coincides with the minimum value of (2.1).

Proof:

- (i) Writing $h_{t+1} = (h_t, u_t, x_{t+1})$, the right-hand side of (2.5) can be written as

$$\min_{\mu_t(u_t|h_t)} \sum_{u_t \in \mathcal{U}} \rho_t(u_t) \mu_t(u_t|h_t) \quad (2.8)$$

where

$$\rho_t(u_t) = C_t(x_t, u_t) + \sum_{x_{t+1} \in \mathcal{X}} J_{t+1}(h_{t+1}) \tau_t(x_{t+1}|x_t, u_t) \quad (2.9)$$

can be regarded as a function of $u_t \in \mathcal{U}$ for each fixed h_t . Since $\mathcal{U}$ is finite, there exists an element $u_t^* \in \mathcal{U}$ such that attains the minimum of $\rho_t(u_t)$:

$$u_t^* \in \arg \min_{u_t} \rho_t(u_t). \quad (2.10)$$

The quantity (2.8) can be minimized by the conditional distribution μ_t^* that chooses u_t^* deterministically:

$$\mu_t^*(u_t|h_t) = \begin{cases} 1 & \text{if } u_t = u_t^* \\ 0 & \text{otherwise.} \end{cases} \quad (2.11)$$

Hence, the existence of μ_t^* is shown by construction.

- (ii) Recall that $J_t^{\mu^*}(h_t)$ is computed by (2.4) with the terminal condition (2.3) by setting $\mu = \mu^*$. Because of the identical terminal condition, the claim (2.6) holds for $t = T$. To complete the proof by backward induction, assume that (2.6) holds for the time step $t + 1$, i.e.,

$$J_{t+1}(h_{t+1}) = J_{t+1}^{\mu^*}(h_{t+1}) \quad (2.12)$$

for all h_{t+1} . Then, we have the following chain of equalities:

$$J_t^{\mu^*}(h_t) = \sum_{u_t} \left[C_t(x_t, u_t) + \sum_{x_{t+1}} J_{t+1}^{\mu^*}(h_{t+1}) \tau_t(x_{t+1}|x_t, u_t) \right] \mu_t^*(u_t|h_t) \quad (2.13a)$$

$$= \sum_{u_t} \left[C_t(x_t, u_t) + \sum_{x_{t+1}} J_{t+1}(h_{t+1}) \tau_t(x_{t+1}|x_t, u_t) \right] \mu_t^*(u_t|h_t) \quad (2.13b)$$

$$= \min_{\mu_t(u_t|h_t)} \sum_{u_t} \left[C_t(x_t, u_t) + \sum_{x_{t+1}} J_{t+1}(h_{t+1}) \tau_t(x_{t+1}|x_t, u_t) \right] \mu_t(u_t|h_t) \quad (2.13c)$$

$$= J_t(h_t). \quad (2.13d)$$

The first equality (2.13a) holds by definition (2.4). The induction hypothesis (2.12) is used in step (2.13b). The equality (2.13c) follows from the fact that μ_t^* is the minimizer of the objective function (c.f., part (i)). The final step (2.13d) follows from the definition (2.5) of $J_t(h_t)$. This completes the proof by induction.

- (iii) The following chain of equalities shows that (2.7) holds for $t = T - 1$:

$$\begin{aligned} & J_{T-1}^{\mu^*}(h_{T-1}) \\ &= \sum_{u_{T-1}} \left[C_{T-1}(x_{T-1}, u_{T-1}) + \sum_{x_T} J_T^{\mu^*}(h_T) \tau_{T-1}(x_T|x_{T-1}, u_{T-1}) \right] \mu_{T-1}^*(u_{T-1}|h_{T-1}) \end{aligned} \quad (2.14a)$$

$$= \min_{\mu_{T-1}} \sum_{u_{T-1}} \left[C_{T-1}(x_{T-1}, u_{T-1}) + \sum_{x_T} J_T^{\mu^*}(h_T) \tau_{T-1}(x_T|x_{T-1}, u_{T-1}) \right] \mu_{T-1}(u_{T-1}|h_{T-1}) \quad (2.14b)$$

$$= \min_{\mu_{T-1}} \sum_{u_{T-1}} \left[C_{T-1}(x_{T-1}, u_{T-1}) + \sum_{x_T} C_T^{\text{exit}}(x_T) \tau_{T-1}(x_T|x_{T-1}, u_{T-1}) \right] \mu_{T-1}(u_{T-1}|h_{T-1}) \quad (2.14c)$$

$$= \min_{\mu_{T-1}} \mathbb{E}^{\mu_{T-1}} [C_{T-1}(x_{T-1}, u_{T-1}) + C_T^{\text{exit}}(x_T)]. \quad (2.14d)$$

The first line (2.14a) is due to the definition (2.4), and the second line (2.14b) is due to the optimality of μ_{T-1}^* . The terminal condition (2.3) is used in step (2.14c). The expression (2.14d) is obtained by rewriting (2.14c) using the expectation symbol $\mathbb{E}[\cdot]$.

To complete the proof by backward induction, assume that (2.7) holds for the time step $t = k + 1$, i.e.,

$$J_{k+1}^{\mu^*}(h_{k+1}) = \min_{\mu_{k+1:T-1}} \mathbb{E}^{\mu_{k+1:T-1}} \left[\sum_{\ell=k+1}^{T-1} C_\ell(x_\ell, u_\ell) + C_T^{\text{exit}}(x_T) \right]. \quad (2.15)$$

Then,

$$J_k^{\mu^*}(h_k) = \sum_{u_k} \left[C_k(x_k, u_k) + \sum_{x_{k+1}} J_{k+1}^{\mu^*}(h_{k+1}) \tau_k(x_{k+1}|x_k, u_k) \right] \mu_k^*(u_k|h_k) \quad (2.16a)$$

$$= \min_{\mu_k} \sum_{u_k} \left[C_k(x_k, u_k) + \sum_{x_{k+1}} J_{k+1}^{\mu^*}(h_{k+1}) \tau_k(x_{k+1}|x_k, u_k) \right] \mu_k(u_k|h_k) \quad (2.16b)$$

$$= \min_{\mu_k} \mathbb{E}^{\mu_k} \left[C_k(x_k, u_k) + J_{k+1}^{\mu^*}(h_{k+1}) \right] \quad (2.16c)$$

$$= \min_{\mu_k} \mathbb{E}^{\mu_k} \left[C_k(x_k, u_k) + \min_{\mu_{k+1:T-1}} \mathbb{E}^{\mu_{k+1:T-1}} \left\{ \sum_{\ell=k+1}^{T-1} C_\ell(x_\ell, u_\ell) + C_T^{\text{exit}}(x_T) \right\} \right] \quad (2.16d)$$

$$= \min_{\mu_{k:T-1}} \mathbb{E}^{\mu_{k:T-1}} \left[\sum_{\ell=k}^{T-1} C_\ell(x_\ell, u_\ell) + C_T^{\text{exit}}(x_T) \right]. \quad (2.16e)$$

The first two equalities follow from the definition (2.4) and the fact that μ_k^* attains the minimum of (2.16b). The induction hypothesis (2.15) is used in step (2.16d). This completes the proof.

(iv) This statement follows from part (iii). ■

Theorem 2.1 implies that an optimal policy can be constructed by solving the backward recursion (2.5) and, for each t and h_t , finding a solution $\mu_t^*(u_t|h_t)$ that attains the minimum in (2.5). Satisfying these conditions is sufficient for the policy $\mu_{0:T-1}^*$ to be optimal, as claimed in part (iv) of Theorem 2.1.

Remark 2.1 *By inspecting the proof of part (i), we observe that the minimum in equation (2.5) is always attained by a deterministic policy. This observation simplifies (2.5) to*

$$J_T(h_T) = C_T^{\text{exit}}(x_T) \quad \forall h_T, \text{ and} \quad (2.17a)$$

$$J_t(h_t) = \min_{u_t} \left[C_t(x_t, u_t) + \sum_{x_{t+1} \in \mathcal{X}} J_{t+1}(h_{t+1}) \tau_t(x_{t+1}|x_t, u_t) \right] \quad \forall h_t, \quad (2.17b)$$

for $t = T-1, T-2, \dots, 0$.

However, even after this simplification, (2.17) still requires the computation of $J_t(h_t)$ for all possible histories h_t . This is clearly inconvenient for practical implementation. Fortunately, the Markov property of the state-space model $\tau_t(x_{t+1}|x_t, u_t)$ and the stage-additive structure of the cost functional significantly relax this requirement.

Theorem 2.2 *Consider the fully observable stochastic optimal control problem (2.1). If both $\mathcal{X}$ and $\mathcal{U}$ are finite spaces, then the following statements hold:*

- (i) *There exists a deterministic and Markov optimal policy of the form $u_t = \mu_t(x_t)$ for all $t = 0, 1, \dots, T-1$, where $\mu_t : \mathcal{X} \rightarrow \mathcal{U}$ is an appropriate function.*

(ii) The optimal cost-to-go function $J_t(h_t)$ depends on $h_t = (x_0, u_0, x_1, u_1, \dots, u_{t-1}, x_t)$ only through x_t . Consequently, one may write $J_t(h_t)$ as $J_t(x_t)$. The function $J_t(x_t)$ satisfies the backward recursion

$$J_t(x_t) = \min_{u_t} \left\{ C_t(x_t, u_t) + \sum_{x_{t+1}} J_{t+1}(x_{t+1}) \tau_t(x_{t+1} | x_t, u_t) \right\} \quad (2.18)$$

for all $t = 0, 1, \dots, T-1$ and $x_t \in \mathcal{X}$ with the terminal condition $J_T(x_T) = C_T^{\text{exit}}(x_T)$.

Proof: We will prove both statements by backward induction. To show (i) and (ii) for $t = T-1$, notice that the optimality condition (2.5) at this time step is

$$J_{T-1}(h_{T-1}) = \min_{\mu_{T-1}} \sum_{u_{T-1}} \rho_{T-1}(x_{T-1}, u_{T-1}) \mu_{T-1}(u_{T-1} | h_{T-1}) \quad (2.19)$$

where

$$\rho_{T-1}(x_{T-1}, u_{T-1}) = C_{T-1}(x_{T-1}, u_{T-1}) + \sum_{x_T} C_T^{\text{exit}}(x_T) \tau_{T-1}(x_T | x_{T-1}, u_{T-1}).$$

The function $\rho_{T-1}(x_{T-1}, u_{T-1})$ depends on h_{T-1} only through x_{T-1} . Moreover, because of the finiteness of $\mathcal{U}$, there exists a minimizer $u_{T-1}^* = \arg \min_{u_{T-1}} \rho_{T-1}(x_{T-1}, u_{T-1})$ for each x_{T-1} . Since u_{T-1}^* only depends on x_{T-1} , one can write it as²

$$u_{T-1}^* = \mu_{T-1}(x_{T-1}). \quad (2.20)$$

Since a deterministic policy minimizes the right-hand side of (2.19), (2.19) simplifies to

$$J_{T-1}(x_{T-1}) = \min_{u_{T-1}} \rho_{T-1}(x_{T-1}, u_{T-1}). \quad (2.21)$$

Therefore, we have shown the existence of a deterministic Markov optimal policy (2.20) and derived the simplified optimality condition (2.21) for $t = T-1$.

Let us now assume that statement (ii) holds for $t = k+1$ for some $0 \leq k \leq T-2$. Since this implies that $J_{k+1}(h_{k+1})$ can be written as $J_{k+1}(x_{k+1})$, the optimality condition (2.5) becomes

$$J_k(h_k) = \min_{\mu_k} \sum_{u_k} \rho_k(x_k, u_k) \mu_k(u_k | h_k). \quad (2.22)$$

where

$$\rho_k(x_k, u_k) = C_k(x_k, u_k) + \sum_{x_{k+1}} J_{k+1}(x_{k+1}) \tau_k(x_{k+1} | x_k, u_k).$$

As in the case with $t = T-1$, the right-hand side of (2.22) is minimized by a deterministic Markov policy

$$u_k^* = \mu_k(x_k), \quad (2.23)$$

which only depends on x_k . Therefore, (2.22) can be simplified to

$$J_k(x_k) = \min_{u_k} \rho_k(x_k, u_k). \quad (2.24)$$

Therefore, we have shown the existence of a deterministic Markov optimal policy (2.23) and derived the simplified optimality condition (2.24) for $t = k$. This completes the proof. ■

Algorithm 1: Dynamic Programming (Finite MDP case)**Input:** Finite-horizon MDP with finite $\mathcal{X}$ and $\mathcal{U}$

- 1 Set $J_T(x) = C_T^{\text{exit}}(x)$ for all $x \in \mathcal{X}$;
- 2 **for** $t = T - 1, T - 2, \dots, 0$ **do**
- 3 | Set $J_t(x) = \min_u \{C_t(x, u) + \sum_{x'} J_{t+1}(x')\tau(x'|x, u)\}$ for all $x \in \mathcal{X}$;
- 4 **end**

Output: The value function $J_t(x)$ for all $t = 0, 1, \dots, T$ and $x \in \mathcal{X}$

Theorem 2.2 guarantees the correctness of the *dynamic programming algorithm* summarized in Algorithm 1. The key formula in this algorithm is equation (2.18), which is commonly referred to as the *dynamic programming equation* or the *Bellman equation*.

A significant advantage of Algorithm 1 is that the optimal cost-to-go function is represented as a function of (x, t) , rather than (h_t, t) as in the earlier recursion (2.17). The cost-to-go function computed by Algorithm 1 is referred to as the *value function*.

The value function is computed backward in time and can be used as a lookup table during policy execution. Specifically, for each state-time pair (x, t) , the optimal control u_t is determined by:

$$u_t \in \arg \min_u \left\{ C_t(x, u) + \sum_{x'} J_{t+1}(x')\tau_t(x'|x, u) \right\}. \quad (2.25)$$

Exercise 2.1 Consider a finite MDP with $|\mathcal{X}| = N$ and $|\mathcal{U}| = M$. How many multiplications are required to complete the backward recursion (2.17)? *Hint:* Notice that MN multiplications are required to evaluate the right-hand side of (2.17b). How many are required for Algorithm 1?

Example 2.1 (*Machine replacement*) The machine replacement problem is a commonly used benchmark for finite-state MDPs. The state space represents the condition of the machine:

$$\mathcal{X} = \{1 \text{ (worst quality)}, \dots, N \text{ (best quality)}\}.$$

At each time step, the user can either replace the machine with a brand-new one or continue using the existing one:

$$\mathcal{U} = \{1 \text{ (replace machine)}, 2 \text{ (use existing machine)}\}.$$

If $u_t = 2$ is chosen, the machine's quality degrades by one unit with probability $\theta \in [0, 1]$. Hence, the transition probability $P_{ij}(k) = P(x_{t+1} = j \mid x_t = i, u_t = k)$ is represented, for $N = 3$, by the matrices

$$P(1) = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad P(2) = \begin{bmatrix} 1 & 0 & 0 \\ \theta & 1 - \theta & 0 \\ 0 & \theta & 1 - \theta \end{bmatrix}.$$

Replacing the machine costs R regardless of its current state, i.e., $C(x = i, 1) = R$ for all $i \in \mathcal{X}$. Continuing to use the existing machine also incurs a cost, which increases as the machine's condition worsens. For example, $C(x = i, 2) = N - i$ for all $i \in \mathcal{X}$. Table 2.1 shows the optimal state-feedback policy obtained by Algorithm 1 when $N = 3$, $T = 15$, $R = 3.5$, $\theta = 0.4$, and $C_T^{\text{exit}}(\cdot) = 0$.

²If there are multiple minimizers, the choice is arbitrary.

Step t	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
$\mu(x=1)$	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2
$\mu(x=2)$	2	2	1	2	2	1	2	2	1	2	2	1	2	2	2
$\mu(x=3)$	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2

Table 2.1: Optimal machine replacement policy.

Exercise 2.2 Solve the machine replacement problem with $N = 10$, $R = 15$, and $T = 15$, while keeping other parameters the same. Obtain the optimal policy and show it in a look-up table similar to Table 2.1.

2.2.2 Continuous state and action spaces

Bellman's optimality principle can be generalized to a broader class of stochastic optimal control problems with continuous $\mathcal{X}$ and $\mathcal{U}$.

Continuous/infinite action spaces

When the space $\mathcal{U}$ is not finite, the existence of an optimal solution to (2.1) is not guaranteed in general. In this case, we are interested in computing³

$$J = \inf_{\mu_{0:T-1}} \mathbb{E}^{\mu_{0:T-1}} \left\{ \sum_{t=0}^{T-1} C_t(x_t, u_t) + C_T^{\text{exit}}(x_T) \right\}. \quad (2.26)$$

We also hope to be able to compute an ϵ -optimal policy $\mu_{0:T-1}^\epsilon$ for an arbitrary parameter $\epsilon > 0$, which satisfies

$$J + \epsilon > \mathbb{E}^{\mu_{0:T-1}^\epsilon} \left\{ \sum_{t=0}^{T-1} C_t(x_t, u_t) + C_T^{\text{exit}}(x_T) \right\}. \quad (2.27)$$

Exercise 2.3 Suppose $\mathcal{X}$ is finite. Show that the infimum (2.26) can be computed by $J = J_0(x_0)$, where $J_0(x_0)$ is obtained by the following modified Bellman equation:

$$J_T(x) = C_T^{\text{exit}}(x) \quad \forall x \in \mathcal{X} \quad (2.28a)$$

$$J_t(x) = \inf_u \left[C_t(x, u) + \sum_{x' \in \mathcal{X}} J_{t+1}(x') \tau_t(x'|x, u) \right] \quad \forall x \in \mathcal{X}, t = T-1, \dots, 1, 0. \quad (2.28b)$$

Describe a procedure to construct an ϵ -optimal policy.

Continuous/infinite state spaces

If $\mathcal{X}$ is not finite, the summation in (2.28b) must be replaced with an integral as follows:

$$J_t(x) = \inf_u \left[C_t(x, u) + \int_{x' \in \mathcal{X}} J_{t+1}(x') \tau_t(dx'|x, u) \right] \quad \forall x \in \mathcal{X}. \quad (2.29)$$

However, dynamic programming based on (2.29) is generally not computationally tractable for two main reasons: (i) storing the values of $J_t(x)$ over the continuum $x \in \mathcal{X}$ requires infinite memory, and (ii) evaluating the integral in (2.29) is typically challenging.

³We use $\inf$ instead of $\min$ if the existence of the minimizer is not guaranteed.

A standard resolution to this difficulty is to introduce a finite set of representative states (also called *features*). Dynamic programming can then be performed exclusively over this reduced feature space, with the hope that the resulting value function closely approximates the true value function characterized by (2.29). This idea, known as *approximate dynamic programming*, forms the foundation for many practical reinforcement learning methods.

An important special case of fully observable continuous-state stochastic optimal control problems for which an efficient solution algorithm exists is the *Linear Quadratic Regulator* problem:

Exercise 2.4 Consider the discrete-time, fully observable state space model

$$x_{t+1} = A_t x_t + B_t u_t + w_t, \quad w_t \sim \mathcal{N}(0, W_t) \quad (2.30)$$

over $t = 0, 1, \dots, T-1$ with the initial state $x_0 \in \mathbb{R}^n$. Consider

$$\min_{\mu_{0:T-1}} \mathbb{E}^{\mu_{0:T-1}} \left[\sum_{t=0}^{T-1} (\|x_t\|_{Q_t}^2 + \|u_t\|_{R_t}^2) + \|x_T\|_{Q_T}^2 \right] \quad (2.31)$$

where $Q_t \succeq 0$ and $R_t \succ 0$. (Note: $\|x\|_Q^2 := x^\top Q x$.)

(i) Show that the Bellman equation (2.29) is equivalent to the Riccati difference equation:

$$S_t = A_t^\top S_{t+1} A_t + Q_t - A_t^\top S_{t+1} B_t (B_t^\top S_{t+1} B_t + R_t)^{-1} B_t^\top S_{t+1} A_t \quad (2.32)$$

which needs to be solved backward in time $t = T-1, T-2, \dots, 0$ with the terminal condition $S_T = Q_T$. Hint: Verify that the value function can be expressed as $J_t(x_t) = \|x_t\|_{S_t}^2 + \sum_{k=t}^{T-1} \text{Tr}(W_k S_{k+1})$.

(ii) Show that the optimal state feedback policy exists and is given by $u_t = K_t x_t$ where the matrix K_t is obtained by

$$K_t = -(B_t^\top S_{t+1} B_t + R_t)^{-1} B_t^\top S_{t+1} A_t. \quad (2.33)$$

2.2.3 Infinite horizon problems

Variable end-time problems

Consider the variable end-time stochastic optimal control problem:

$$\min_{\mu_{0:T_f-1}} \mathbb{E}^{\mu_{0:T_f-1}} \left[\sum_{t=0}^{T_f-1} C(x_t, u_t) + C^{\text{exit}}(x_{T_f}) \right] \quad (2.34)$$

where $T_f = \min\{t > 0 | x_t \in \mathcal{X}_{\text{abs}}\}$. For simplicity, we assume that both $\mathcal{X}$ and $\mathcal{U}$ are finite. The well-posedness of problem (2.34) requires the boundedness of the cost function, which is guaranteed under certain assumptions. For example, the following condition is sufficient to guarantee that T_f is finite with probability one and that (2.34) is bounded.

Assumption 2.1 There exists an integer m such that there is a positive probability that an absorbing state will be reached after no more than m steps, regardless of the policy used and the initial state. That is, for each $\mu_{0:m}$ and $x_0 \in \mathcal{X}$, there exists $0 \leq t \leq m$ such that

$$P\{x_t \in \mathcal{X}_{\text{abs}} | x_0, \mu_{0:m}\} > 0. \quad (2.35)$$

Algorithm 2: Dynamic Programming (Finite MDP, variable end-time)**Input:** MDP with variable end-time, iteration number K

- 1 Initialize $J_0(x) = C_T^{\text{exit}}(x)$ for $x \in \mathcal{X}_{\text{abs}}$;
 - 2 Initialize $J_0(x)$ arbitrarily for $x \in \mathcal{X} \setminus \mathcal{X}_{\text{abs}}$;
 - 3 **for** $k = 0, 1, \dots, K$ (“until convergence”) **do**
 - 4 $J_{k+1}(x) = C_T^{\text{exit}}(x)$ for $x \in \mathcal{X}_{\text{abs}}$;
 - 5 $J_{k+1}(x) = \min_u \{C(x, u) + \sum_{x'} J_k(x')\tau(x'|x, u)\}$ for $x \in \mathcal{X} \setminus \mathcal{X}_{\text{abs}}$;
 - 6 **end**
- Output:** $J_K(x)$ for $x \in \mathcal{X} \setminus \mathcal{X}_{\text{abs}}$

Due to the infinite-horizon nature of (2.34) necessitates an infinite iteration of the dynamic programming operations in general. Specifically, consider Algorithm 2.

The following result [5] guarantees the convergence of Algorithm 2:

Theorem 2.3 *Under Assumption 2.1, the following statements hold for the sequence J_k generated by Algorithm 2:*

- (i) *For each $x \in \mathcal{X} \setminus \mathcal{X}_{\text{abs}}$, $J_k(x)$ converges to $J^*(x)$, where J^* is the unique solution to*

$$J(x) = C_T^{\text{exit}}(x), \quad \forall x \in \mathcal{X}_{\text{abs}} \quad (2.36a)$$

$$J(x) = \min_{u \in \mathcal{U}} \left\{ C(x, u) + \sum_{x'} J(x')\tau(x'|x, u) \right\}, \quad \forall x \in \mathcal{X} \setminus \mathcal{X}_{\text{abs}}. \quad (2.36b)$$

- (ii) *The unique solution J^* to (2.36) is the value function for (2.34). That is,*

$$J^*(x) = \min_{\mu_t: T_f-1} \mathbb{E}^{\mu_t: T_f-1} \left\{ \sum_{k=t}^{T_f-1} C(x_k, u_k) + C^{\text{exit}}(x_{T_f}) \right\} \quad (2.37)$$

for each $t > 0$ and $x \in \mathcal{X} \setminus \mathcal{X}_{\text{abs}}$, where $T_f = \min\{k > t | x_k \in \mathcal{X}_{\text{abs}}\}$.

- (iii) *A stationary (time-invariant) policy $\mu^* : \mathcal{X} \setminus \mathcal{X}_{\text{abs}} \rightarrow \mathcal{U}$ is optimal if and only if*

$$\mu^*(x) \in \arg \min_{u \in \mathcal{U}} \left\{ C(x, u) + \sum_{x' \in \mathcal{X}} J^*(x')\tau(x'|x, u) \right\} \quad (2.38)$$

for each $x \in \mathcal{X} \setminus \mathcal{X}_{\text{abs}}$.

Discounted cost

Let us now consider how the dynamic programming algorithm is applicable to discounted problems:

$$\min_{\mu_{0:\infty}} \sum_{t=0}^{\infty} \gamma^t \mathbb{E}^{\mu_{0:\infty}} [C(x_t, u_t)]. \quad (2.39)$$

where $\gamma \in (0, 1)$ is a discount factor. Unlike the previous case with variable end-times, each episode in the discounted problem has an infinite length. Despite this difference, it is not difficult to see that (2.39) can be rewritten as an equivalent *undiscounted, variable end-time* problem if one notices that (2.39) can be reinterpreted as the problem of minimizing the undiscounted cost

when the process terminates with probability $1 - \gamma$ at each time step (regardless of the current state). Specifically, introduce an artificial absorbing state x_{abs} and define a new state space by $\bar{\mathcal{X}} = \mathcal{X} \cup \{x_{\text{abs}}\}$. On this new state space, define a new state transition kernel by

$$\bar{\tau}(x'|x, u) = \gamma\tau(x'|x, u), \quad \forall x' \in \mathcal{X}, x \in \mathcal{X}, u \in \mathcal{U} \quad (2.40a)$$

$$\bar{\tau}(x_{\text{abs}}|x, u) = 1 - \gamma, \quad x \in \mathcal{X}, u \in \mathcal{U} \quad (2.40b)$$

$$\bar{\tau}(x'|x_{\text{abs}}, u) = 0, \quad \forall x' \in \mathcal{X}, u \in \mathcal{U} \quad (2.40c)$$

$$\bar{\tau}(x_{\text{abs}}|x_{\text{abs}}, u) = 1, \quad \forall u \in \mathcal{U}. \quad (2.40d)$$

Problem (2.39) is equivalent to the following variable end-time problem with the modified state transition kernel (2.40):

$$\min_{\mu_{0:T_f-1}} \mathbb{E}^{\mu_{0:T_f-1}} \left\{ \sum_{t=0}^{\infty} C(x_t, u_t) \right\} \quad (2.41)$$

with zero exit cost, i.e., $C^{\text{exit}}(\cdot) = 0$. This equivalence allows us to reuse Algorithm 2. The dynamic programming algorithm for (2.41) is shown in Algorithm 3.

Algorithm 3: Dynamic Programming (Finite MDP, discounted cost)

Input: MDP with discounted cost, iteration number K

- 1 Initialize $J_0(x)$ arbitrarily for $x \in \mathcal{X} \setminus \mathcal{X}_{\text{abs}}$;
- 2 **for** $k = 0, 1, \dots, K$ (“until convergence”) **do**
- 3 | $J_{k+1}(x) = \min_u \{C(x, u) + \sum_{x' \in \mathcal{X}} J_k(x') \bar{\tau}(x'|x, u)\}$ for $x \in \mathcal{X} \setminus \mathcal{X}_{\text{abs}}$;
- 4 **end**

Output: $J_K(x)$ for $x \in \mathcal{X}$

Example 2.2 (*Machine replacement*) Recall the machine replacement problem (Example 2.1). This time, consider an infinite-horizon version of the same problem with a discount factor $\gamma = 0.9$. The result of Algorithm 3 with the initial condition $J_0 = [0 \ 0 \ 0]$ is shown in Figure 2.2. The values of J_k converge asymptotically to their steady-state values. At steady-state, the corresponding optimal policy is $\mu(x = 1) = 1$ (Replace), $\mu(x = 2) = 2$ (Use), and $\mu(x = 3) = 2$ (Use).

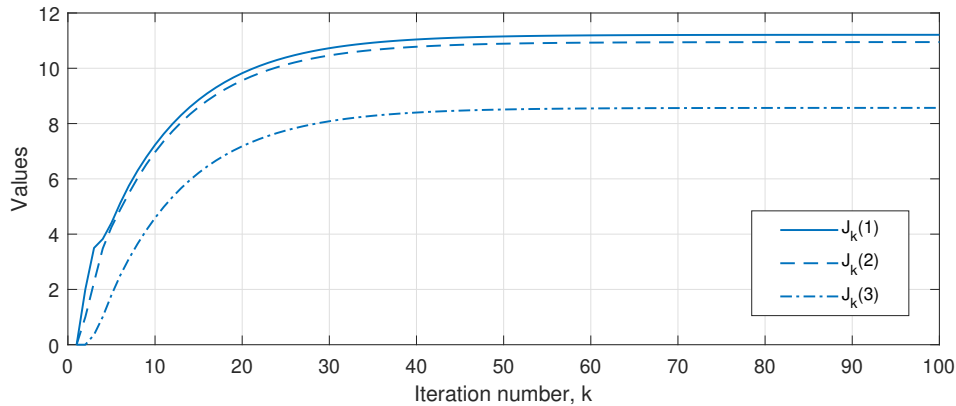


Figure 2.1: Algorithm 3 for the discounted machine replacement problem.

2.3 State Estimation

So far, our focus has been on stochastic optimal control problems with fully observable states. For such problems, we have shown (Theorem 2.2) that an optimal policy can always be chosen as a static state feedback policy (i.e., a deterministic and Markov policy). In this section, we shift our focus to stochastic optimal control problems with partially observable states. Clearly, state feedback is not an admissible policy in such scenarios. The key technique that bridges the gap between fully observable and partially observable stochastic control problems is the recursive state estimation algorithm known as *filtering*.

A filtering algorithm allows us to compute, at least approximately, the posterior distribution $p(x_t|y_{0:t})$ for each possible realization of the observed sequence $y_{0:t}$. Since recalculating $p(x_t|y_{0:t})$ from scratch at every time step is computationally cumbersome, it is generally desirable to compute $p(x_t|y_{0:t})$ by recursively updating the previous posterior $p(x_{t-1}|y_{0:t-1})$ using the most recent observation y_t . A formula that enables such recursive updates is referred to as *Bayesian filtering*, since the update rule is derived from Bayes' rule.⁴

Remark 2.2 Often, what is meant by optimal estimation is the computation of the conditional mean $\hat{x}_t = \mathbb{E}[x_t|y_{0:t}]$. This is the most sensible choice when the goal is to minimize the mean-square estimation error $\mathbb{E}[(x_t - \hat{x}_t)^\top (x_t - \hat{x}_t)]$. For this reason, $\hat{x}_t = \mathbb{E}[x_t | y_{0:t}]$ is also called the minimum mean-square error (MMSE) estimate.⁵ Another commonly used concept is the maximum a posteriori (MAP) estimate $\hat{x}_t = \arg \max_{x \in \mathcal{X}} p(x_t|y_{0:t})$. The Viterbi algorithm is a well-known MAP estimator that is widely used in error-correcting codes for wireless communication [7].

Neither the MMSE nor the MAP estimator fully characterizes the observer's knowledge. For example, the MMSE estimate alone does not convey the degree of certainty in the estimation. When the posterior distribution $p(x_t|y_{0:t})$ is known to be Gaussian, the pair $(\hat{x}_t, P_{t|t})$ of the MMSE estimate $\hat{x}_t$ and the error covariance $P_{t|t}$ fully characterizes the observer's knowledge. In such cases, we say that $(\hat{x}_t, P_{t|t})$ is a *sufficient statistic* for $p(x_t|y_{0:t})$. In general, however, there is no finite-dimensional sufficient statistic for Bayesian filtering.

2.3.1 Bayesian Filtering

Consider a partially observable Markov process with a state transition kernel $\tau(x_{t+1}|x_t)$ and an observation kernel $o(y_t|x_t)$.⁶ For simplicity, we first assume that there is no control input. We will later see that incorporating control inputs is a straightforward generalization.

⁴Suppose $p(x, y)$ denotes a joint probability mass function (for discrete random variables) or a joint probability density function (for continuous random variables), and assume that $p(x, y) \neq 0$. Bayes' rule states that $p(x | y) = \frac{p(x, y)}{p(y)} = \frac{p(y|x)p(x)}{\int_{\mathcal{X}} p(y|x)p(x) dx}$.

⁵More generally, the conditional mean minimizes a loss function that can be expressed as the expected *Bregman divergence* [6].

⁶For notational simplicity, the Borel measurable transition and observation kernels $\tau_t(dx_{t+1}|x_t)$ and $o(dy_t|x_t)$ will be denoted as $\tau_t(x_{t+1}|x_t)$ and $o(y_t|x_t)$, and will be interpreted as conditional PMFs (for discrete cases) and conditional PDFs (for continuous cases). In applications where conditional PMFs/PDFs do not exist (e.g., hybrid systems), one should invoke the Borel model (Section 1.1.1) to generalize the derivation below. We will also suppress the subscripts and write τ_t and o_t as τ and o , but these kernels can be time-varying in the following discussions.

Assume that the initial state distribution $\pi_0 := p(x_0)$ is known. Our goal is to compute the posterior distribution $\pi_t := p(x_t|y_{0:t})$ at each time step t . Here, π_t denotes a probability distribution over $\mathcal{X}_t$, referred to as the *belief*. Note that π_t depends on the realization of the observed measurements $y_{0:t}$. We are interested in how to recursively update the belief π_{t-1} to obtain π_t , given the new measurement y_t . If $p(y_{0:t}) \neq 0$, we have⁷

$$\pi_t(x_t) = p(x_t|y_{0:t}) = \frac{p(x_t, y_{0:t})}{p(y_{0:t})} = \frac{p(x_t, y_{0:t})}{\int_{\mathcal{X}_t} p(x_t, y_{0:t}) dx_t}. \quad (2.42)$$

The joint distribution $p(x_t, y_{0:t})$ is expressed as

$$p(x_t, y_{0:t}) = p(y_t|x_t, y_{0:t-1}) p(x_t, y_{0:t-1}) \quad (2.43a)$$

$$= o(y_t|x_t) \int_{\mathcal{X}_{t-1}} p(x_t, x_{t-1}, y_{0:t-1}) dx_{t-1} \quad (2.43b)$$

$$= o(y_t|x_t) \int_{\mathcal{X}_{t-1}} p(x_t|x_{t-1}, y_{0:t-1}) p(x_{t-1}, y_{0:t-1}) dx_{t-1} \quad (2.43c)$$

$$= o(y_t|x_t) \int_{\mathcal{X}_{t-1}} \tau(x_t|x_{t-1}) \underbrace{p(x_{t-1}|y_{0:t-1})}_{\pi_{t-1}(x_{t-1})} p(y_{0:t-1}) dx_{t-1}. \quad (2.43d)$$

In step (2.43b), we used the identity $p(y_t|x_t, y_{0:t-1}) = o(y_t|x_t)$, which holds because y_t is conditionally independent of $y_{0:t-1}$ given x_t (by the observation model). Similarly, in step (2.43d), we used $p(x_t|x_{t-1}, y_{0:t-1}) = \tau(x_t|x_{t-1})$, due to the conditional independence implied by the state transition model.

Substituting (2.43) into (2.42), we obtain the recursive formula:

$$\pi_t(x_t) = \frac{o(y_t|x_t) \int_{\mathcal{X}_{t-1}} \tau(x_t|x_{t-1}) \pi_{t-1}(x_{t-1}) dx_{t-1}}{\int_{\mathcal{X}_t} o(y_t|x_t) \int_{\mathcal{X}_{t-1}} \tau(x_t|x_{t-1}) \pi_{t-1}(x_{t-1}) dx_{t-1} dx_t}. \quad (2.44)$$

Equation (2.44) is known as the *Bayesian filter*. It describes how the belief state π_{t-1} is updated to π_t upon receiving the new measurement y_t . All terms on the right-hand side of (2.44) are known: $\tau(x_t|x_{t-1})$ and $o(y_t|x_t)$ are the state transition and observation kernels, respectively, and π_{t-1} is the belief state of the previous step.

Remark 2.3 The denominator of (2.44) serves as a normalization factor, ensuring that $\int_{\mathcal{X}_t} \pi_t(x_t) dx_t = 1$ at each time step. If we omit the normalization, we obtain the unnormalized Bayesian filter:

$$\tilde{\pi}_t(x_t) = o(y_t|x_t) \int_{\mathcal{X}_{t-1}} \tau(x_t|x_{t-1}) \tilde{\pi}_{t-1}(x_{t-1}) dx_{t-1}. \quad (2.45)$$

Equation (2.45) has the advantage of being linear in the unnormalized belief states $\tilde{\pi}_t$, which makes it computationally more attractive. While $\int_{\mathcal{X}_t} \tilde{\pi}_t(x_t) \neq 1$ in general, the identity $\tilde{\pi}_t \propto \pi_t$ still holds.

In general, the Bayes filter (2.44) cannot be implemented directly, as the integral expressions involved are often intractable. There are, however, two important special cases of (2.44) that admit tractable implementations. These are known as *Hidden Markov Model (HMM) filtering* and *Kalman filtering*, which we summarize next.

⁷For notational simplicity, we write $\int_{\mathcal{X}_t}$ instead of $\int_{x_t \in \mathcal{X}_t}$.

2.3.2 Hidden Markov Model (HMM) filter

The first special case in which (2.44) is tractable is when both $\mathcal{X}$ and $\mathcal{Y}$ are finite spaces. The belief π_t is represented as an $|\mathcal{X}|$ -dimensional vector in this case. By replacing the integrals in (2.44) with summations, we obtain the following HMM filter:

$$\pi_t(x_t) = \frac{o(y_t|x_t) \sum_{x_{t-1} \in \mathcal{X}} \tau(x_t|x_{t-1}) \pi_{t-1}(x_{t-1})}{\sum_{x_t \in \mathcal{X}} o(y_t|x_t) \sum_{x_{t-1} \in \mathcal{X}} \tau(x_t|x_{t-1}) \pi_{t-1}(x_{t-1})}. \quad (2.46)$$

Example 2.3 (*Machine replacement*) Recall that the optimal time-invariant policy for the discounted machine replacement problem (Example 2.2) is

$$\mu(x=1) = 1 \text{ (Replace)}, \quad \mu(x=2) = 2 \text{ (Use)}, \quad \mu(x=3) = 2 \text{ (Use)}.$$

Applying this policy yields the autonomous Markov process shown in Figure 2.2, governed by the state transition kernel $\tau(x_{t+1}|x_t)$:

$$\begin{bmatrix} \tau(1|1) & \tau(2|1) & \tau(3|1) \\ \tau(1|2) & \tau(2|2) & \tau(3|2) \\ \tau(1|3) & \tau(2|3) & \tau(3|3) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ \theta & 1-\theta & 0 \\ 0 & \theta & 1-\theta \end{bmatrix}. \quad (2.47)$$

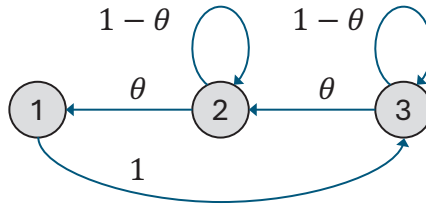


Figure 2.2: Machine replacement process.

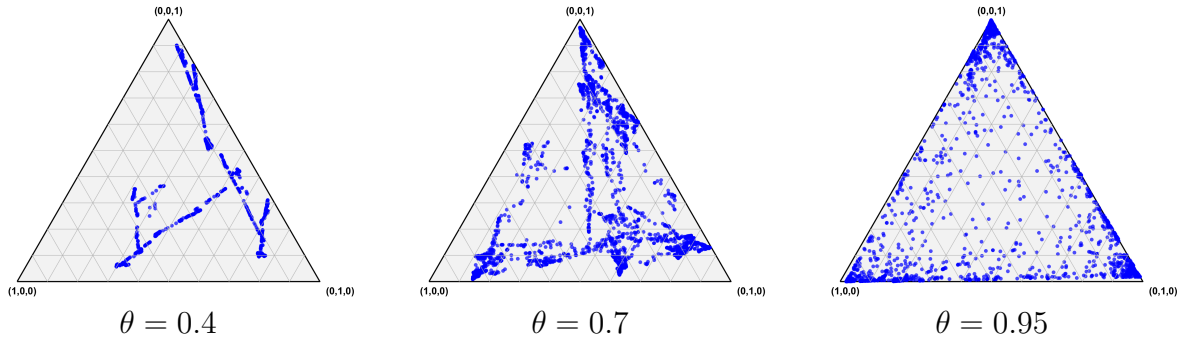
Suppose this Markov process (2.47) is observed by an external agent who can only estimate the current state x_t through a noisy observation y_t . Assume the observation kernel $o(y_t|x_t)$ is given by

$$\begin{bmatrix} o(1|1) & o(2|1) & o(3|1) \\ o(1|2) & o(2|2) & o(3|2) \\ o(1|3) & o(2|3) & o(3|3) \end{bmatrix} = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.1 & 0.7 \end{bmatrix}. \quad (2.48)$$

Suppose the initial belief is $\pi_0 = (1/3, 1/3, 1/3)$, and the agent begins observing y_t at $t = 1$. Table 2.2 shows a simulated sequence of (x_t, y_t) , and the corresponding beliefs π_t computed using the HMM filter. To visualize the evolution of the belief states, Figure 2.3 shows the first 1000 samples of π_t plotted on a ternary diagram, where the space $\Delta(\mathcal{X})$ of probability distributions over $\mathcal{X}$ is represented as a triangle. When $\theta = 0.4$, beliefs are concentrated around several small regions in $\Delta(\mathcal{X})$. However, when $\theta = 0.95$, beliefs are spread across the entire simplex.

t	x_t	y_t	$\pi_t(1)$	$\pi_t(2)$	$\pi_t(3)$
0	—	—	0.3333	0.3333	0.3333
1	1	1	0.4375	0.3125	0.2500
2	3	3	0.0259	0.1194	0.8545
3	3	2	0.0261	0.6789	0.2948
4	2	1	0.6026	0.3330	0.0643
5	2	2	0.0917	0.4663	0.4418
6	2	3	0.0518	0.2537	0.6943
7	2	2	0.0545	0.6935	0.2518
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Table 2.2: Simulated Markov process (2.47) with the observation kernel (2.48) and the HMM filter.

Figure 2.3: First 1000 samples of the belief π_t obtained by the HMM filter.

2.3.3 Kalman filter

The second special case is when $\mathcal{X} = \mathbb{R}^n$ and $\mathcal{Y} = \mathbb{R}^m$, and the state transition kernel $\tau(x_{t+1}|x_t)$ and the observation kernel $o(y_t|x_t)$ are given by

$$x_{t+1} = A_t x_t + w_t, \quad w_t \sim \mathcal{N}(0, W_t) \quad (2.49a)$$

$$y_t = C_t x_t + v_t, \quad v_t \sim \mathcal{N}(0, V_t) \quad (2.49b)$$

for $t = 0, 1, 2, \dots$. We assume that $W_t \succeq 0$ (positive semidefinite) and $V_t \succ 0$ (positive definite). It can be shown that if the initial belief π_0 is Gaussian, then π_t remains Gaussian for all $t = 0, 1, 2, \dots$. (See Exercise 2.5 below.) Since the mean and covariance form a sufficient statistic for Gaussian distributions, to implement the Bayesian filter, it is enough to keep track of:

$$\hat{x}_{t|t} := \mathbb{E}[x_t | y_{0:t}] \quad \text{and} \quad (2.50a)$$

$$P_{t|t} := \mathbb{E}[(x_t - \hat{x}_{t|t})(x_t - \hat{x}_{t|t})^\top] \quad (2.50b)$$

so that we have $\pi_t \sim \mathcal{N}(\hat{x}_{t|t}, P_{t|t})$. To implement the Kalman filter, it is also convenient to introduce

$$\hat{x}_{t|t-1} := \mathbb{E}[x_t | y_{0:t-1}] \quad \text{and} \quad (2.51a)$$

$$P_{t|t-1} := \mathbb{E}[(x_t - \hat{x}_{t|t-1})(x_t - \hat{x}_{t|t-1})^\top] \quad (2.51b)$$

and divide the operations within each time step into the prediction and update phases. Assuming that the initial belief $\pi_0 \sim \mathcal{N}(\hat{x}_{0|0}, P_{0|0})$ is given, the Kalman filtering algorithm is described as follows:

Prediction:

$$\hat{x}_{t|t-1} = A_t \hat{x}_{t-1|t-1} \quad (2.52a)$$

$$P_{t|t-1} = A_t P_{t-1|t-1} A_t^\top + W_t \quad (2.52b)$$

Update:

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + P_{t|t-1} C_t^\top (C_t P_{t|t-1} C_t^\top + V_t)^{-1} (y_t - C_t \hat{x}_{t|t-1}) \quad (2.53a)$$

$$P_{t|t} = P_{t|t-1} - P_{t|t-1} C_t^\top (C_t P_{t|t-1} C_t^\top + V_t)^{-1} C_t P_{t|t-1}. \quad (2.53b)$$

The procedure is summarized in Algorithm 4. See also Figure 2.4.

Algorithm 4: Kalman filter.

Input: State space model (2.49), initial belief $\pi_0 \sim \mathcal{N}(\hat{x}_{0|0}, P_{0|0})$

1 **for** $k = 1, 2, \dots, K$ **do**

2 **Prediction:** Compute $(\hat{x}_{t|t-1}, P_{t|t-1})$ by (2.52);

3 **Update:** Compute $(\hat{x}_{t|t}, P_{t|t})$ by (2.53);

4 **end**

Output: Belief sequence $\pi_t \sim \mathcal{N}(\hat{x}_{t|t}, P_{t|t})$ for $t = 1, 2, \dots$

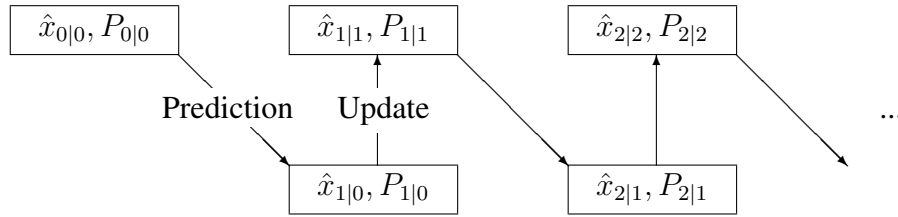


Figure 2.4: Prediction and update steps in Kalman filtering.

Remark 2.4 It can be shown that $\eta_t := y_t - C_t \hat{x}_{t|t-1}$, $t = 0, 1, 2, \dots$, called the *innovation process*, is a sequence of zero-mean independent random variables, i.e., $\mathbb{E} \eta_s \eta_t^\top = 0$ for all $0 \leq s < t$. Due to this property, the Kalman filter is sometimes called the *whitening filter*.

Remark 2.5 Suppose that the process noise w_t and the measurement noise v_t are individually independent and zero-mean random processes with covariance matrices W_t and V_t respectively, but they are not Gaussian random vectors. In this case, $\hat{x}_{t|t}$ computed by Algorithm 4 is no longer the MMSE estimate, and the Kalman filter is no longer a Bayesian filter. Nevertheless, it can be shown that the Kalman filter is still the best linear MMSE estimator, and $P_{t|t-1}$ and $P_{t|t}$ are still correct values of the estimation error covariances.

Exercise 2.5 Let (x, y) be jointly Gaussian random vectors such that

$$\begin{bmatrix} x \\ y \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \bar{x} \\ \bar{y} \end{bmatrix}, \begin{bmatrix} P_{xx} & P_{xy} \\ P_{yx} & P_{yy} \end{bmatrix} \right), \quad P_{yy} \succ 0. \quad (2.54)$$

(i) Show that $p(x|y) \sim \mathcal{N}(\hat{x}, P_{x|y})$ where $\hat{x} = \bar{x} + P_{xy}P_{yy}^{-1}(y - \bar{y})$ and $P_{x|y} = P_{xx} - P_{xy}P_{yy}^{-1}P_{yx}$.

(ii) Derive the Kalman filter formula (2.52) and (2.53). Specifically, first show that if $p(x_{t-1}|y_{0:t-1}) \sim \mathcal{N}(\hat{x}_{t-1|t-1}, P_{t-1|t-1})$, then $p(x_t|y_{0:t-1}) \sim \mathcal{N}(\hat{x}_{t|t-1}, P_{t|t-1})$ where $\hat{x}_{t|t-1}$ and $P_{t|t-1}$ are given by (2.52). Then, using the result of part (i), show that $p(x_t|y_{0:t}) \sim \mathcal{N}(\hat{x}_{t|t}, P_{t|t})$ where $\hat{x}_{t|t}$ and $P_{t|t}$ are given by (2.53).

2.4 Optimal Output Feedback Synthesis

Let us consider a partially observable state-space model $\tau(x_{t+1}|x_t, u_t)$ with the observation kernel $o(y_t|x_t)$, and consider the finite-horizon stochastic optimal control problem:

$$\min_{\mu_{0:T-1}} \mathbb{E}^{\mu_{0:T-1}} \left\{ \sum_{t=0}^{T-1} C_t(x_t, u_t) + C_T^{\text{exit}}(x_T) \right\}. \quad (2.55)$$

This is a minimization problem over the space of history-dependent randomized policies $\mu_{0:T-1} = \{\mu_t(u_t|y_{0:t}, u_{0:t-1})\}_{t=0}^{T-1}$.

2.4.1 Belief State Representation

Let $\pi_t(x_t) = p(x_t|y_{0:t}, u_{0:t-1})$ denote the belief. As in the previous section, π_t represents the conditional distribution over the state space $\mathcal{X}$ given the information $(y_{0:t}, u_{0:t-1})$ available to the decision-making agent at time step t . The only difference here is that x_t is a controlled Markov process with input u_t . However, this difference poses no difficulty in generalizing the Bayesian filter (2.44), which becomes:

$$\pi_{t+1}(x_{t+1}) = \frac{o(y_{t+1}|x_{t+1}) \int_{\mathcal{X}_t} \tau(x_{t+1}|x_t, u_t) \pi_t(x_t) dx_t}{\int_{\mathcal{X}_{t+1}} o(y_{t+1}|x_{t+1}) \int_{\mathcal{X}_t} \tau(x_{t+1}|x_t, u_t) \pi_t(x_t) dx_t dx_{t+1}}. \quad (2.56)$$

The central idea of the belief state representation is that (2.56) can be viewed as a state in its own right in which π_t serves as a state variable (called the *belief state*). To emphasize this point, (2.56) may be written as

$$\pi_{t+1} = T(\pi_t, u_t, y_{t+1}) \quad (2.57)$$

from which it is apparent that the transition of the belief state $\pi_t \rightarrow \pi_{t+1}$ is determined by the control input u_t and the new observation y_{t+1} . Although u_t is known to the decision maker at time t , y_{t+1} is not. Thus, we can interpret (2.57) as a stochastic control system in which y_{t+1} plays the role of process noise.

This viewpoint becomes especially valuable once we observe that the belief state π_t itself is a controlled Markov process with control input u_t . In particular, we state the following:

Proposition 2.1 *Let π_t , $t = 0, 1, 2, \dots$ be the sequence of belief states computed via (2.56). Then π_{t+1} depends on $(\pi_{0:t}, u_{0:t}, y_{0:t})$ only through (π_t, u_t) . That is,*

$$p(\pi_{t+1} | \pi_{0:t}, u_{0:t}, y_{0:t}) = p(\pi_{t+1} | \pi_t, u_t). \quad (2.58)$$

Proof: If (π_t, u_t) is known, then additional knowledge of y_{t+1} fully determines π_{t+1} via (2.57). Therefore, it suffices to show that the random variable y_{t+1} depends on $(\pi_{0:t}, u_{0:t}, y_{0:t})$ only through (π_t, u_t) . This can be verified as follows:

$$p(y_{t+1} | \pi_{0:t}, u_{0:t}, y_{0:t}) = \int_{\mathcal{X}_{t+1}} \int_{\mathcal{X}_t} p(y_{t+1}, x_{t+1}, x_t | \pi_{0:t}, u_{0:t}, y_{0:t}) dx_t dx_{t+1} \quad (2.59a)$$

$$= \int_{\mathcal{X}_{t+1}} \int_{\mathcal{X}_t} o(y_{t+1} | x_{t+1}) \tau(x_{t+1} | x_t, u_t) p(x_t | \pi_t) dx_t dx_{t+1} \quad (2.59b)$$

$$= p(y_{t+1} | \pi_t, u_t). \quad (2.59c)$$

The simplification follows from:

- the observation model: $p(y_{t+1} | x_{t+1}, x_t, \pi_{0:t}, u_{0:t}, y_{0:t}) = o(y_{t+1} | x_{t+1})$;
- the transition model: $p(x_{t+1} | x_t, \pi_{0:t}, u_{0:t}, y_{0:t}) = \tau(x_{t+1} | x_t, u_t)$;
- the belief property: $p(x_t | \pi_{0:t}, u_{0:t}, y_{0:t}) = p(x_t | \pi_t)$.

■

The conditional measure $p(\pi_{t+1} | \pi_t, u_t)$ in Proposition 2.1 will also be written as $\tau(\pi_{t+1} | \pi_t, u_t)$ in the sequel to emphasize that it serves as a state transition kernel.

With Proposition 2.1 in hand, consider the partially observable stochastic optimal control problem (2.55). Can we express its objective function in terms of the belief state π_t , rather than the unobservable state x_t ? The answer is affirmative and can be verified as follows:

$$\begin{aligned} & \mathbb{E} \left\{ \sum_{t=0}^{T-1} C_t(x_t, u_t) + C_T^{\text{exit}}(x_T) \right\} \\ &= \mathbb{E} \left\{ \sum_{t=0}^{T-1} \mathbb{E} [C_t(x_t, u_t) | u_{0:t-1}, y_{0:t}] + \mathbb{E} [C_T^{\text{exit}}(x_T) | u_{0:T-1}, y_{0:T}] \right\} \end{aligned} \quad (2.60a)$$

$$= \mathbb{E} \left\{ \sum_{t=0}^{T-1} \int_{\mathcal{X}_t} C_t(x_t, u_t) \underbrace{p(x_t | u_{0:t-1}, y_{0:t-1})}_{\pi_t(x_t)} dx_t + \int_{\mathcal{X}_T} C_T^{\text{exit}}(x_T) \underbrace{p(x_T | u_{0:T-1}, y_{0:T-1})}_{\pi_T(x_T)} dx_T \right\} \quad (2.60b)$$

$$= \mathbb{E} \left\{ \sum_{t=0}^{T-1} \tilde{C}_t(\pi_t, u_t) + \tilde{C}_T^{\text{exit}}(\pi_T) \right\}. \quad (2.60c)$$

Step (2.60a) uses the law of iterated expectations. The expression (2.60b) can be rewritten as (2.60c) by introducing new functions:

$$\tilde{C}_t(\pi_t, u_t) := \int_{\mathcal{X}_t} C_t(x_t, u_t) \pi_t(x_t) dx_t, \quad t = 0, 1, \dots, T-1, \quad (2.61a)$$

$$\tilde{C}_T^{\text{exit}}(\pi_T) := \int_{\mathcal{X}_T} C_T^{\text{exit}}(x_T) \pi_T(x_T) dx_T. \quad (2.61b)$$

To summarize, we have reformulated the original problem (2.55), defined over the state-space model $\tau(x_{t+1}|x_t, u_t)$ and $o(y_t|x_t)$, into the equivalent problem:

$$\min_{\mu_{0:T-1}} \mathbb{E}^{\mu_{0:T-1}} \left\{ \sum_{t=0}^{T-1} \tilde{C}_t(\pi_t, u_t) + \tilde{C}_T^{\text{exit}}(\pi_T) \right\}, \quad (2.62)$$

subject to the belief-state dynamics $\tau(\pi_{t+1}|\pi_t, u_t)$.

A major benefit of this reformulation is that the belief state π_t is always fully observable, even though the true state x_t is not.⁸ Since (2.62) is now a state feedback synthesis problem, the search for an optimal policy can be restricted, without loss of generality, to the class of static state feedback (i.e., deterministic and Markov) policies:

$$\mu_t : \Delta(\mathcal{X}) \rightarrow \mathcal{U}, \quad t = 0, 1, \dots, T-1, \quad (2.63)$$

where $\Delta(\mathcal{X})$ denotes the *belief space*, i.e., the space of probability measures on the Borel σ -algebra $\mathcal{B}_{\mathcal{X}}$. This reformulation enables the application of a dynamic programming algorithm over the belief space.

Algorithm 5: Dynamic Programming over belief space (finite-horizon case)

Input: Belief state representation (2.62) of a finite-horizon partially observable stochastic optimal control problem

- 1 Set $J_T(\pi_T) = \tilde{C}_T^{\text{exit}}(\pi_T)$, $\forall \pi \in \Delta(\mathcal{X})$;
- 2 **for** $t = T-1, T-2, \dots, 0$ **do**
- 3 Set $J_t(\pi_t) =$
 $\min_{u_t \in \mathcal{U}} \left\{ \tilde{C}_t(\pi_t, u_t) + \int_{\pi_{t+1} \in \Delta(\mathcal{X})} J_{t+1}(\pi_{t+1}) \tau(\pi_{t+1}|\pi_t, u_t) d\pi_{t+1} \right\}$, $\forall \pi_t \in \Delta(\mathcal{X})$;
- 4 Set $\mu_t^*(\pi_t) =$
 $\arg \min_{u_t \in \mathcal{U}} \left\{ \tilde{C}_t(\pi_t, u_t) + \int_{\pi_{t+1} \in \Delta(\mathcal{X})} J_{t+1}(\pi_{t+1}) \tau(\pi_{t+1}|\pi_t, u_t) d\pi_{t+1} \right\}$, $\forall \pi_t \in \Delta(\mathcal{X})$;
- 5 **end**

Output: The value function $J_t(\cdot)$ and the optimal policy $\mu_t^*(\cdot)$ for all $t = 0, 1, \dots, T$

The optimal policy $\mu_{0:T-1}^*$ for the partially observable stochastic optimal control problem is obtained by Algorithm 5.⁹ However, Algorithm 5 cannot be implemented directly in practice because both $J_t(\cdot)$ and $\mu_t^*(\cdot)$ must be evaluated over the continuous simplex $\Delta(\mathcal{X})$. As we saw in Example 2.3, the belief state π_t can take any value in $\Delta(\mathcal{X})$.

Observe that the belief-state transition kernel $\tau(\pi_{t+1}|\pi_t, u_t)$ is given by Proposition 2.1. If it is more convenient to work with $p(y_{t+1}|\pi_t, u_t)$, Bellman's equation in Algorithm 5 can be written as

$$J_t(\pi_t) = \min_{u_t \in \mathcal{U}} \left\{ \tilde{C}_t(\pi_t, u_t) + \int_{\mathcal{Y}_{t+1}} J_{t+1}(T(\pi_t, u_t, y_{t+1})) p(y_{t+1}|\pi_t, u_t) dy_{t+1} \right\}, \quad (2.64)$$

$$\mu_t^*(\pi_t) = \arg \min_{u_t \in \mathcal{U}} \left\{ \tilde{C}_t(\pi_t, u_t) + \int_{\mathcal{Y}_{t+1}} J_{t+1}(T(\pi_t, u_t, y_{t+1})) p(y_{t+1}|\pi_t, u_t) dy_{t+1} \right\}. \quad (2.65)$$

⁸It is reasonable to assume that any “self-aware” agent knows its own belief.

⁹In Algorithm 5, we assume that the minimum in Bellman's equation is attained.

2.4.2 Special case 1: POMDP

The stochastic optimal control problem (2.55) is commonly referred to as a *partially observable Markov decision process (POMDP)* when $\mathcal{X}$, $\mathcal{U}$, and $\mathcal{Y}$ are finite. Recall that under this setting the HMM filter and the (fully observable) MDP are each computationally tractable. However, their joint use in the belief-space formulation leads to an intractable control-synthesis problem. This is because the belief space $\Delta(\mathcal{X})$ is continuous even when $\mathcal{X}$ is finite, and the value function update in Bellman's equation (Algorithm 5) must still be performed over all $\pi_t \in \Delta(\mathcal{X})$.

Exercise 2.6 Suppose $\mathcal{X}$, $\mathcal{U}$, and $\mathcal{Y}$ are finite, and consider the finite-horizon POMDP. Show that $J_t(\pi_t)$ is piecewise linear and concave in π_t for each $t = 0, 1, \dots, T$.

2.4.3 Special case 2: LQG Control

Consider the discrete-time dynamics

$$x_{t+1} = A_t x_t + B_t u_t + w_t, \quad w_t \sim \mathcal{N}(0, W_t) \quad (2.66a)$$

$$y_t = C_t x_t + v_t, \quad v_t \sim \mathcal{N}(0, V_t) \quad (2.66b)$$

for $t = 0, 1, \dots, T-1$ with the initial condition $x_0 \sim \mathcal{N}(\hat{x}_{0|-1}, P_{0|-1})$. The process noise w_t and the measurement noise v_t are mutually independent with positive semidefinite covariance matrices W_t and V_t , and are temporally independent. Introducing $Q_t \succeq 0$ and $R_t \succ 0$ for each t , the discrete-time LQG control problem over a finite horizon is formulated as follows:

$$\min \sum_{t=0}^{T-1} \mathbb{E} (\|x_t\|_{Q_t}^2 + \|u_t\|_{R_t}^2) + \mathbb{E} \|x_T\|_{Q_T}^2 \quad (2.67)$$

where the domain of optimization is over the space of output feedback policies. The following proposition gives a complete solution to (2.67).

Proposition 2.2 The optimal policy $\{\mu_t(u_t|y_{0:t}, u_{0:t-1})\}_{t=0}^{T-1}$ for (2.67) is given by

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + L_t(y_t - C_t \hat{x}_{t|t-1}) \quad (2.68a)$$

$$u_t = K_t \hat{x}_{t|t} \quad (2.68b)$$

$$\hat{x}_{t+1|t} = A_t \hat{x}_{t|t} + B_t u_t \quad (2.68c)$$

for $t = 0, 1, \dots, T-1$, and the optimal value of (2.67) is

$$\text{Tr}(S_0 P_{0|0}) + \sum_{t=0}^{T-1} [\text{Tr}(W_t S_{t+1}) + \text{Tr}(\Theta_t P_{t|t})]. \quad (2.69)$$

Here, the matrices $\{P_{t|t}, L_t, P_{t+1|t}\}_{t=0}^{T-1}$ are computed by the following forward recursion:

$$P_{t|t} = (P_{t|t-1}^{-1} + C_t^\top V_t^{-1} C_t)^{-1} \quad (2.70a)$$

$$L_t = P_{t|t-1} C_t^\top (C_t P_{t|t-1} C_t^\top + V_t)^{-1} \quad (2.70b)$$

$$P_{t+1|t} = A_t P_{t|t} A_t^\top + W_t. \quad (2.70c)$$

The matrices $\{S_t, M_t, K_t, \Theta_t\}_{t=0}^{T-1}$ are computed by the following backward recursion:

$$S_t = A_t^\top S_{t+1} A_t - A_t^\top S_{t+1} B_t M_t^{-1} B_t^\top S_{t+1} A_t + Q_t \quad (2.71a)$$

$$M_t = B_t^\top S_{t+1} B_t + R_t \quad (2.71b)$$

$$K_t = -M_t^{-1} B_t^\top S_{t+1} A_t \quad (2.71c)$$

$$\Theta_t = K_t^\top M_t K_t \quad (2.71d)$$

with the terminal condition $S_T = Q_T$. As in the general case of partially observable stochastic optimal control problems, the solution architecture presented by Proposition 2.2 can be viewed as a belief state feedback. Thanks to the LQG nature of the problem, the belief state is always a Gaussian distribution specified by mean and covariance, which the Kalman filter can recursively compute. In each time step, the Kalman filtering operation is divided into the *prediction* step and the *update* step:

- Prediction: Given the belief state $\pi_{t-1} := p(x_{t-1}|y_{1:t-1}, u_{1:t-2})$ of the previous time step, the prediction operation computes the *prior* distribution $p(x_t|y_{1:t-1}, u_{1:t-1})$ using the knowledge of the state space model (2.66a). Specifically, the operations (2.68c) and (2.70c) correspond to the prediction step.
- Update: The update step fuses the latest measurement y_t into the prior distribution $p(x_t|y_{1:t-1}, u_{1:t-1})$ to obtain the *posterior* distribution $\pi_t := p(x_t|y_{1:t}, u_{1:t-1})$. Specifically, the operations (2.68a) and (2.70a) correspond to the update step. The observer gain matrix L_t appearing in (2.68a) is called the *Kalman gain* and is computed by (2.70b).

It follows that $\hat{x}_{t|t-1} = \mathbb{E}[x_t|y_{1:t-1}, u_{1:t-1}]$ and $\hat{x}_{t|t} = \mathbb{E}[x_t|y_{1:t}, u_{1:t-1}]$, and they are the minimum mean-square error estimates of x_t given $(y_{1:t-1}, u_{1:t-1})$ and $(y_{1:t}, u_{1:t-1})$, respectively. The corresponding estimation error covariances are $\text{Cov}(x_t - \hat{x}_{t|t-1}) = P_{t|t-1}$ and $\text{Cov}(x_t - \hat{x}_{t|t}) = P_{t|t}$.

Once the posterior state estimate $\hat{x}_{t|t}$ is obtained in each time step, the optimal control input u_t is determined by (2.68b). The control law of this type is commonly called the *certainty equivalence controller*, since it is obtained by replacing the state vector x_t in the optimal LQR law $u_t = K_t x_t$ for fully observable systems¹⁰ with its estimate $\hat{x}_{t|t}$. In other words, the optimality of the state feedback gain matrix K_t obtained by (2.71) is invariant regardless of whether the full state measurement is available or Kalman filtering is used to compute its estimate. The fact that the Kalman gain L_t and the state feedback gain K_t can be designed independently in the case of the LQG control problem is known as the *separation principle*.

Exercise 2.7 Define the value function

$$J_t(\pi_t) := \min_{\mu_{t:T-1}} \sum_{k=t}^{T-1} \mathbb{E}(\|x_k\|_{Q_k}^2 + \|u_k\|_{R_k}^2) + \mathbb{E}\|x_T\|_{Q_T}^2 \quad (2.72)$$

Prove (2.69) by showing that

$$J_t(\pi_t) = \text{Tr}(S_t P_{t|t}) + \sum_{k=t}^{T-1} [\text{Tr}(W_k S_{k+1}) + \text{Tr}(\Theta_k P_{k|k})]. \quad (2.73)$$

for each $t = 0, 1, \dots, T$.

¹⁰Recall Exercise 2.4.

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